

Expt 4 Design of Low Pass Filter using Windowing Techniques

4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

Code;

clc;

clear;

close all;

%% Input Parameters

fp = 1000; % Passband cutoff frequency (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (Even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency

%% FIR Low Pass Filter using Rectangular Window

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('Rectangular Window Coefficients:');

disp(b_rect');

%% FIR Low Pass Filter using Hamming Window

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('Hamming Window Coefficients:');

disp(b_hamm');

%% FIR Low Pass Filter using Hann Window

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b_hann = fir1(N, wc, 'low', hann(N+1));

disp('Hann Window Coefficients:');
disp(b_hann);

%% Frequency Response - Rectangular Window
figure;
freqz(b_rect,1,1024,fs);
grid on;
title('Low-pass FIR Filter using Rectangular Window');

%% Frequency Response - Hamming Window
figure;
freqz(b_hamm,1,1024,fs);
grid on;
title('Low-pass FIR Filter using Hamming Window');

%% Frequency Response - Hann Window
figure;
freqz(b_hann,1,1024,fs);
grid on;
title('Low-pass FIR Filter using Hann Window');

%% Compare All Frequency Responses
figure;

[H1,f] = freqz(b_rect,1,1024,fs);
plot(f,20*log10(abs(H1)), 'LineWidth',1.5);

```

```
hold on;
```

```
[H2,~] = freqz(b_hamm,1,1024,fs);
```

```
plot(f,20*log10(abs(H2)), 'LineWidth',1.5);
```

```
[H3,~] = freqz(b_hann,1,1024,fs);
```

```
plot(f,20*log10(abs(H3)), 'LineWidth',1.5);
```

```
grid on;
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');
```

```
title('Comparison of FIR Low-pass Filters');
```

```
legend('Rectangular','Hamming','Hann');
```

```
axis([0 fs/2 -100 10]);
```

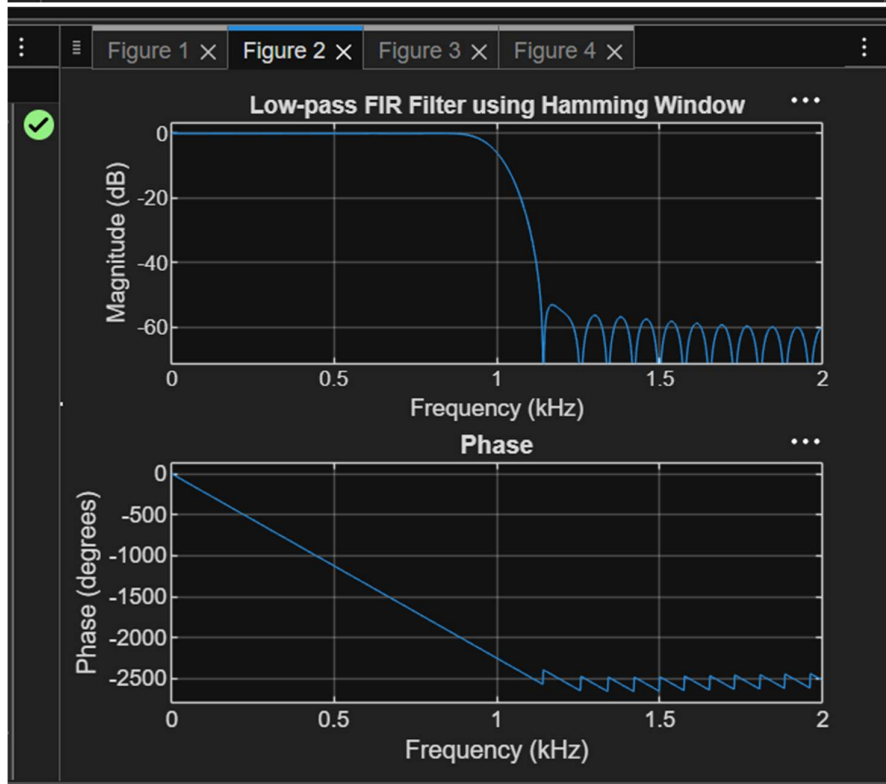
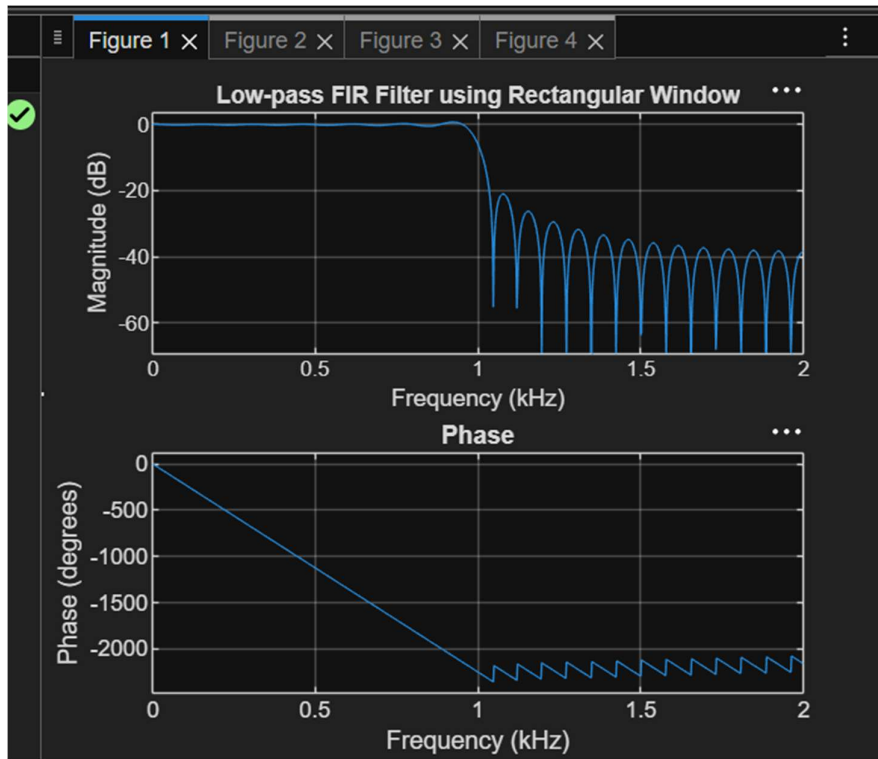
```
%% Display Filter Information
```

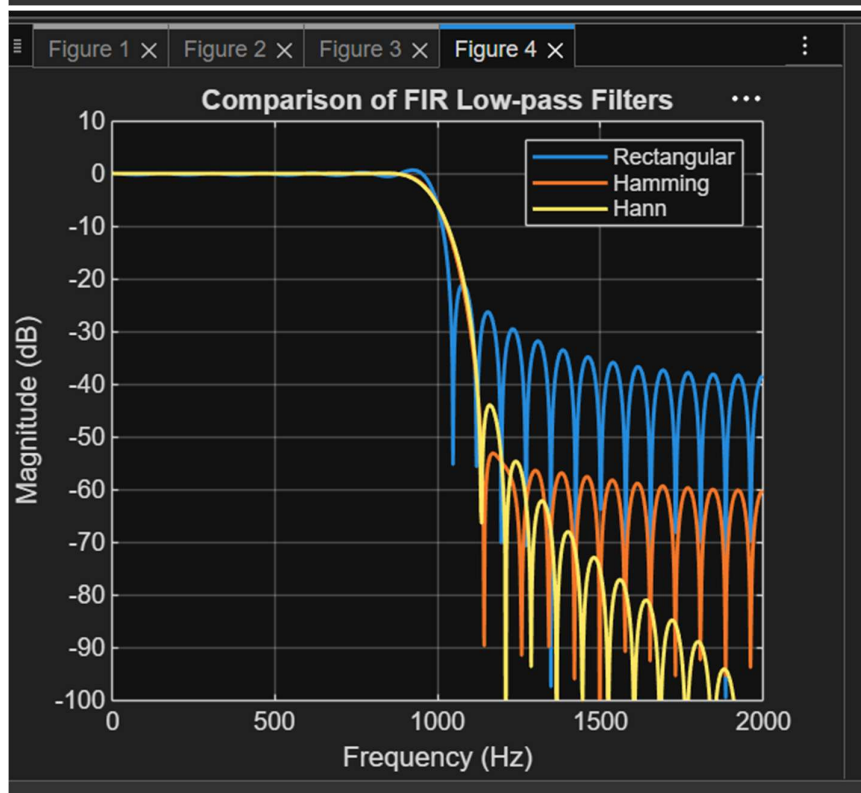
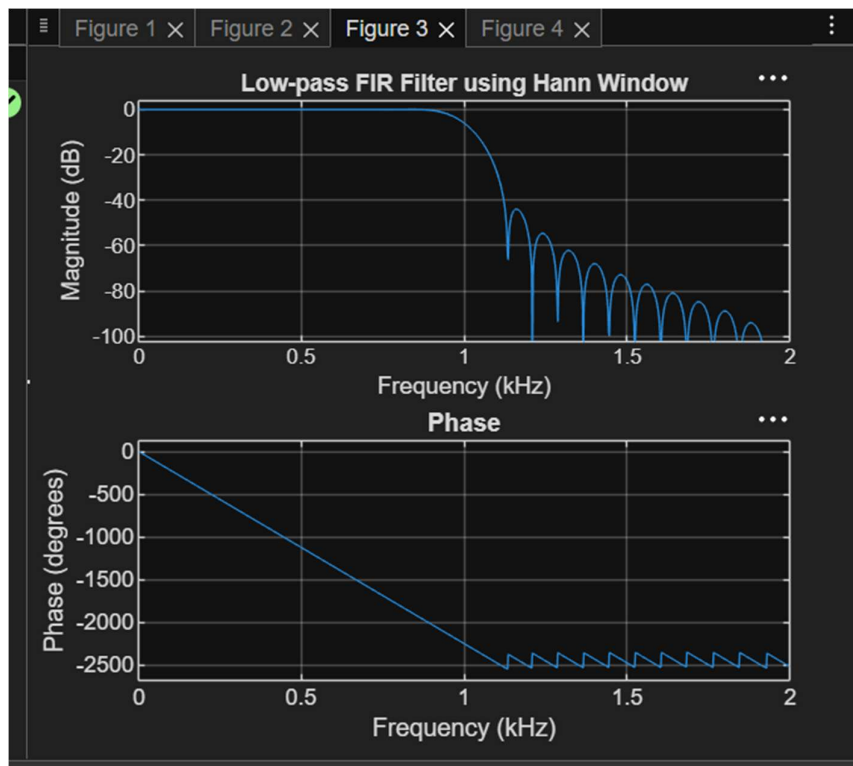
```
fprintf('\nFilter Order      : %d\n', N);
```

```
fprintf('Sampling Frequency    : %d Hz\n', fs);
```

```
fprintf('Cutoff Frequency       : %d Hz\n', fp);
```

```
fprintf('Normalized Cutoff (wc) : %.2f\n', wc);
```





1. An FIR Low-Pass Filter was successfully designed using the Rectangular Window technique for the specified cutoff frequency, sampling frequency, and filter order.